

FINAL EXPLANATION: MATHEMATICS GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: MATHEMATICS GRADE 10 - SUB-STRAND 3.4 LINEAR MOTION

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

Final Explanation: Overtaking on the Naivasha Highway. Driving question: How can we use distance, time, speed, velocity and graphs to describe and predict how one car overtakes another on a Kenyan road? Remember the clip near Naivasha: Wanjiru in the white Subaru (overtaking car B) passes Mr Kamau's blue Toyota Axio (overtaken car A) while Mr Otieno's lorry (oncoming lorry L) approaches from Nakuru. Take the direction towards Nakuru as positive. Answer every section in the space below each prompt. Show your working, write units with every answer, and use evidence from your graphs, your split-time table and your Driving Question Board. Use the words 'overtaking car B', 'overtaken car A' and 'oncoming lorry L', and always say whose viewpoint you are using when you talk about relative speed. Use the table below in Sections 2 to 4.

t (s)	v of B (m/s)	B distance (m)	A distance (m)
0	20	0	0
1	22	21	20
2	24	44	40
3	26	69	60
4	28	96	80
5	30	125	100
6	30	155	120
7	30	185	140
7.5	30	200	150

Section 1: Describing the motion of each vehicle (KICD Question 1: How do you determine the velocity of a moving body?)

Car A covers 150 m in 7.5 s, and the oncoming lorry L travels

(a) Velocity = displacement / time. You find how far the body has changed position in a straight line, in a stated direction, and divide by the time taken. Speed uses total distance (a scalar), but velocity uses displacement and direction (a vector), and the

towards Naivasha, covering 150 m in the same 7.5 s. (a) Explain how you determine the velocity of a moving body, and say how velocity is different from speed. (b) Calculate the velocity of overtaken car A. (c) Convert the speedometer readings of 72 km/h and 108 km/h to m/s. (d) Find the velocity of oncoming lorry L and explain what its sign tells you.	direction is shown by the sign. A speedometer shows speed only. (b) Car A: $v = 150 \text{ m} / 7.5 \text{ s} = 20 \text{ m/s}$ towards Nakuru, so +20 m/s. (c) Divide km/h by 3.6: $72 / 3.6 = 20 \text{ m/s}$ and $108 / 3.6 = 30 \text{ m/s}$. (d) Oncoming lorry L: its speed is $150 \text{ m} / 7.5 \text{ s} = 20 \text{ m/s}$. It travels towards Naivasha, opposite to the positive direction (towards Nakuru), so its velocity is -20 m/s. The negative sign shows direction, not slowing down.
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Section 2: Acceleration from numbers and from a graph (KICD Question 2: How do you determine the acceleration of a moving body?)

Use the table. (a) Explain how you determine the acceleration of a moving body and calculate the acceleration of overtaking car B while its velocity rises from its starting value to its final value. State the time it takes. (b) Sketch the velocity-time graph for car B from $t = 0$ to 7.5 s and for car A on the same axes. Use t (s) on the x-axis and v (m/s) on the y-axis. (c) Explain how the graph also gives the acceleration of car B and what the graph shows for car A. (d) A braking matatu slows from 20 m/s to 0 in 10 s. Find its acceleration and say what the sign means.	(a) Acceleration = change in velocity / time, $a = (v - u)/t$. From the table, B's velocity goes from $u = 20 \text{ m/s}$ to $v = 30 \text{ m/s}$ in $t = 5 \text{ s}$, so $a = (30 - 20)/5 = 2 \text{ m/s}^2$. (b) Car B's graph is a straight line rising from (0, 20) to (5, 30), then a horizontal line at 30 m/s to $t = 7.5 \text{ s}$. Car A's graph is a horizontal line at 20 m/s from 0 to 7.5 s. Axes are labelled t (s) and v (m/s), with 1 cm = 1 s on the x-axis. (c) The gradient of a velocity-time graph is acceleration: $(30 - 20)/5 = 2 \text{ m/s}^2$ for the sloping section of B. A's horizontal line has gradient 0, so car A has zero acceleration and constant velocity. (d) $a = (0 - 20)/10 = -2 \text{ m/s}^2$. The negative sign shows deceleration: the matatu is slowing down, and its deceleration is 2 m/s^2.
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Section 3: How far did each car travel?

(a) Use the area under B's velocity-time graph, split into a trapezium and a rectangle, to find the distance travelled by car B in 7.5 s. (b) Find the distance travelled by car A in	(a) Trapezium (0 to 5 s): $(20 + 30)/2 \times 5 = 125 \text{ m}$. Rectangle (5 to 7.5 s): $30 \times 2.5 = 75 \text{ m}$. Total for B = $125 + 75 = 200 \text{ m}$. (b) Car A: rectangle $20 \times 7.5 = 150 \text{ m}$. (c) The table gives B = 200 m and A = 150 m at $t = 7.5 \text{ s}$, and at $t = 5 \text{ s}$ B = 125 m, which matches the trapezium. The areas agree with the split-time table. (d) $200 - 150 = 50 \text{ m}$. B travels 50 m more than A because it speeds up at 2 m/s^2 while A stays steady. (e) $50 \text{ m} = 20 \text{ m}$ starting gap + 5 m (length of A) + 5 m (length of B) + 20 m finishing gap. At $t = 5 \text{ s}$ B has gained $125 - 100 = 25 \text{ m}$, so B's front is level with A's front.
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the same time using its graph. (c) Check your answers against the table. (d) How much further did B travel than A, and what does this tell you about the overtake? (e) Car B starts 20 m behind A and ends 20 m ahead of A, and each car is 5 m long. Show how the extra distance B gains fits with these lengths and gaps.	
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Section 4: Why does car A look slow from car B?

Both speedometers show high readings, yet car A looks slow from car B's point of view. (a) Find the speed of B relative to A once B has reached its final speed, in m/s and in km/h. (b) What was B's speed relative to A at $t = 0$, and why? (c) Use the B and A distances in the table at $t = 4$ s and $t = 5$ s to show that the gap is growing. (d) Find the closing speed of B and oncoming lorry L in m/s and km/h, stating the viewpoint. (e) Use these ideas to explain how two vehicles that are both moving fast can appear to move slowly relative to each other.	(a) Same direction, so subtract: B relative to A = $30 - 20 = 10$ m/s. $10 \times 3.6 = 36$ km/h. (b) At $t = 0$ both cars are at 20 m/s, so B relative to A = $20 - 20 = 0$ m/s. It grows to 10 m/s at $t = 5$ s as B accelerates. (c) At $t = 4$ s, B minus A = $96 - 80 = 16$ m; at $t = 5$ s, $125 - 100 = 25$ m. B gains 9 m in the fifth second, and the gain per second is increasing because B relative to A is increasing. (d) B and L move in opposite directions, so add: $30 + 20 = 50$ m/s = 180 km/h. From B's viewpoint, the lorry closes at 50 m/s. (e) Relative speed depends on the viewpoint. When two vehicles go the same way you subtract their speeds, so A looks slow from B (only 10 m/s, 36 km/h, and 0 at the start) even though each speedometer reads high. The oncoming lorry looks very fast from B because speeds add when vehicles move towards each other.
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Section 5: Final Explanation - Is the overtake safe, and how do we describe and predict it?

Wanjiru's overtake takes 7.5 s. The lorry L is 500 m ahead of B at $t = 0$. (a) How far does the lorry travel in 7.5 s? (b) How much road do B and L use together, and how far away is the lorry when B pulls back into lane? (c) Using the closing speed, find the safety margin in	(a) Lorry L: $20 \text{ m/s} \times 7.5 \text{ s} = 150 \text{ m}$. (b) B and L together use $200 + 150 = 350 \text{ m}$. When B is back in lane, the lorry is $500 - 350 = 150 \text{ m}$ away. (c) Closing speed is 50 m/s, so margin = $150 / 50 = 3.0 \text{ s}$. A 150 m (3.0 s) margin means the overtake is safe. (d) With the lorry 400 m away: $400 - 350 = 50 \text{ m}$, and $50 / 50 = 1.0 \text{ s}$. This is unsafe because there is very little time to spare. (e) We can describe each vehicle with numbers: velocity = displacement / time gives car A +20 m/s and the lorry -20 m/s (opposite direction), and acceleration $a = (v - u)/t$ gives B 2 m/s^2 as it speeds up from 20 to 30 m/s in 5 s. Velocity-time graphs show these motions: the gradient gives acceleration and the area gives distance (B 200 m, A 150 m, so B gains 50 m in 7.5 s). Relative speed explains what drivers see: B relative to A is only 10 m/s (36 km/h), so A looks slow, while B and L close at 50 m/s (180 km/h). Combining these, B and L need 350 m of road, so with the lorry 500 m away there are 150 m (3.0 s) to spare and the overtake is safe, but with the lorry at 400 m only 1.0 s is left and it is unsafe. So speedometer readings alone do not decide safety; the
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seconds. Is the overtake safe? (d) What if the lorry were only 400 m away at $t = 0$? (e) Write a final answer to the driving question: How can we use distance, time, speed, velocity and graphs to describe and predict how one car overtakes another? Refer to your answers to KICD Questions 1 and 2, graphs, relative speed and the safety decision.	distances, times and relative speeds do.
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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Velocity and acceleration methods (KICD Questions 1 and 2)	States velocity = displacement/time and $a = (v - u)/t$, explains the vector nature and sign of velocity, converts units correctly and calculates A's +20 m/s, the lorry's -20 m/s, B's 2 m/s ² and the matatu's -2 m/s ² with units.	Uses both formulas correctly with minor slips in units or in explaining the sign, and gets most values right.	Confuses speed with velocity or omits conversion, the sign or units, and gets several values wrong.
Graphs: gradient and area	Sketches correct labelled velocity-time graphs for A and B, links gradient to acceleration, and finds B 200 m (125 + 75) and A 150 m from areas, checking against the table.	Draws mostly correct graphs and finds the distances with a small error or without checking against the table.	Graphs are missing labels or shapes are wrong, and the gradient or area is not linked to acceleration or distance.
Relative speed and viewpoint	Subtracts for the same direction (10 m/s, 36 km/h) and adds for opposite directions (50 m/s, 180 km/h), states the viewpoint, notes 0 at $t = 0$ and explains why A looks slow from B.	Calculates relative speeds correctly but gives limited explanation or omits the viewpoint or the starting value.	Adds or subtracts incorrectly, or cannot explain why A looks slow.
Safety decision and use of data	Correctly finds the lorry's 150 m, the 350 m of road used, the 150 m (3.0 s) margin and the unsafe 50 m (1.0 s) case, and uses table values accurately.	Reaches the correct decision with one calculation error or a missing unit or time conversion.	Decision is not supported by calculations or the numbers are inconsistent with the data.
Final Explanation answering the driving question	Connects velocity, acceleration, graphs, relative speed and safety in a clear, logical explanation that uses the Naivasha evidence and answers both KICD questions.	Answers the driving question using most ideas but with some links missing.	Lists facts without linking them to the overtake or the driving question.

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